

Fields

Defⁿ

A field is

a set F equipped with

(Addition)

$$+ : F \times F \longrightarrow F$$
$$(a, b) \longmapsto a + b$$

(Multiplication)

$$\cdot : F \times F \longrightarrow F$$
$$(a, b) \longmapsto a \cdot b$$

(Zero)

$$0 \in F$$

(One)

$$1 \in F$$

Satisfying:

Addition

$$\bullet a + b = b + a$$

$$\bullet a + 0 = 0 + a = a$$

$$\bullet \forall a, \exists -a$$

s.t.

$$a + (-a) = 0$$

$$\bullet (a + b) + c = a + (b + c)$$

Multiplication

$$a \cdot b = b \cdot a$$

$$a \cdot 1 = 1 \cdot a = a$$

$$\bullet \forall a \neq 0, \exists a^{-1} \text{ s.t.}$$

$$a \cdot a^{-1} = a^{-1} \cdot a = 1$$

$$\bullet a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

Together

$$a \cdot (b + c)$$

$$=$$

$$a \cdot b + a \cdot c$$

Distributivity

Commutativity

inverses

associativity

HW 2

Show:

(1) $\forall a \in F, 0 \cdot a = 0$

(2) $\forall a \in F, (-1) \cdot a = -a$

(3) $\forall a, b \in F, \text{ if }$

$a \cdot b = 0$ then

either $a = 0$ or

$b = 0$

Examples of fields

(1) $\mathbb{N}$: We have addition & multiplication, but no inverses for addition or multiplication and also no 0

(2) We can fix the problem of no 0
no inverses for addition

In this way, we get $\mathbb{Z}$. Still no inverses for multiplication

③ Fix this.

In this way, we obtain

$$\mathbb{Q} = \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}$$

$$\frac{m}{n} = m \cdot \frac{1}{n}$$

$\mathbb{Q}$ is a field!!

④ $\mathbb{R}$ is also a field

$\mathbb{Q} \subseteq \mathbb{R}$ is a subfield

(5) $\mathbb{Z}/n\mathbb{Z}$ has addition & multiplication & 0 & 1

They satisfy all the conditions for a field except for inverses for multiplication (e.s. fails in $\mathbb{Z}/4\mathbb{Z}$)

Fact: If $n=p$ is a prime ($p=2, 3, 5$ etc), then we do have inverses for multiplication &

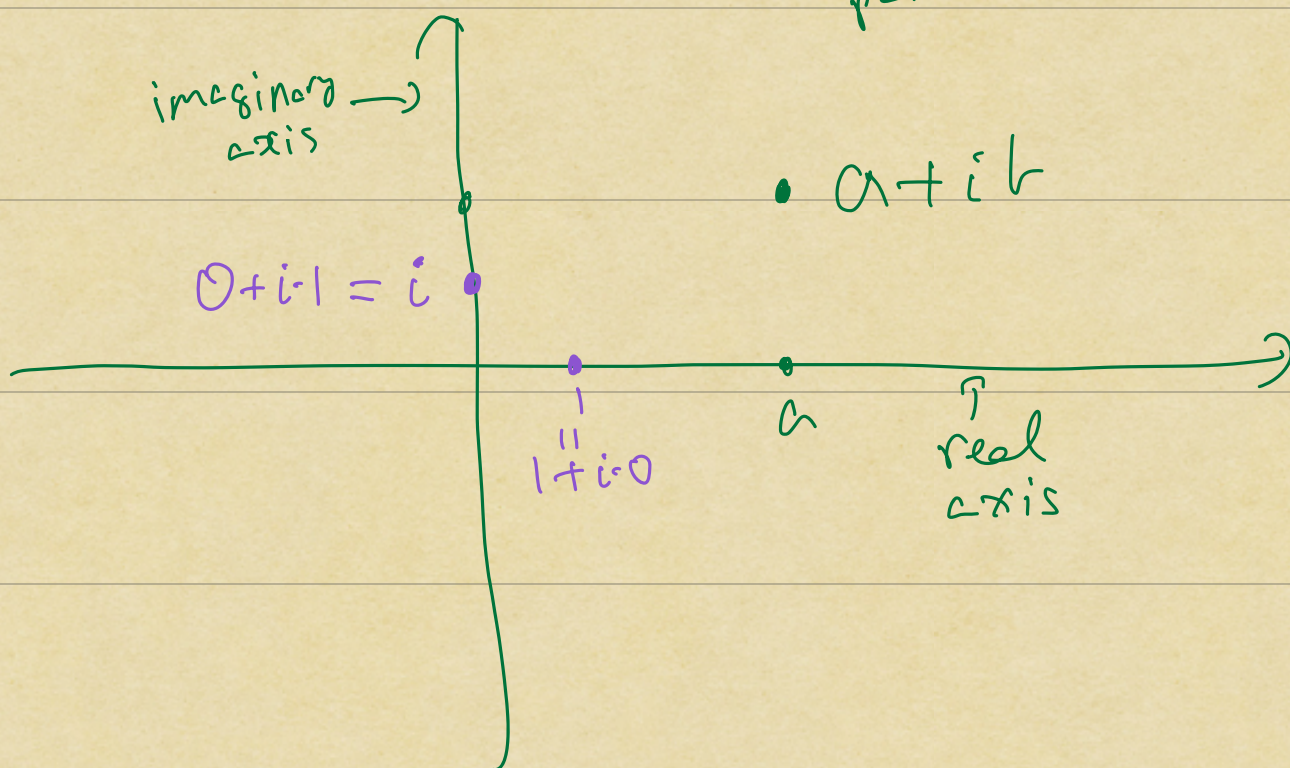
$\mathbb{Z}/p\mathbb{Z}$ is a field

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Complex numbers

$x^2 + 1 = 0$ has no solution in $\mathbb{R}$

$$\mathbb{C} = \{ \underset{\substack{\uparrow \\ \text{real part}}}{a} + i \underset{\substack{\uparrow \\ \text{imaginary part}}}{b} : a, b \in \mathbb{R} \}$$



$\mathbb{R} \subseteq \mathbb{C}$: Complex numbers with zero imaginary part
" $\{ a + i \cdot 0 : a \in \mathbb{R} \}$

The arithmetic of $\mathbb{C}$
is determined by:

- Addition

$$(a + ib) + (c + id) \\ = (a + c) + i(b + d)$$

- Multiplication

$$i^2 = -1$$

$$\Rightarrow (a + ib)(c + id)$$

$$= ac + \underbrace{(ib) \cdot c + a \cdot (id)}_{i(cd + bc)} + \underbrace{(ib)(id)}_{-bd} \\ = (ac - bd) + i(cd + bc)$$

- 0 & 1

$$0 = 0 + i \cdot 0, \quad 1 = 1 + i \cdot 0$$

$$(1+i \cdot 0)(a+i \cdot b) \stackrel{\vee}{=} a+i \cdot b$$

• Inverse for addition

$$-(a+ib) = (-a)+i(-b)$$

• Inverse for multiplication

$$z = a+ib \neq 0$$

$$z \cdot \bar{z} = a^2+b^2 = |z|^2 \neq 0$$

$$\Rightarrow z \cdot \frac{\bar{z}}{|z|^2} = 1$$

$$\Rightarrow z^{-1} = \frac{\bar{z}}{|z|^2}$$

$\mathbb{C}$ is a field

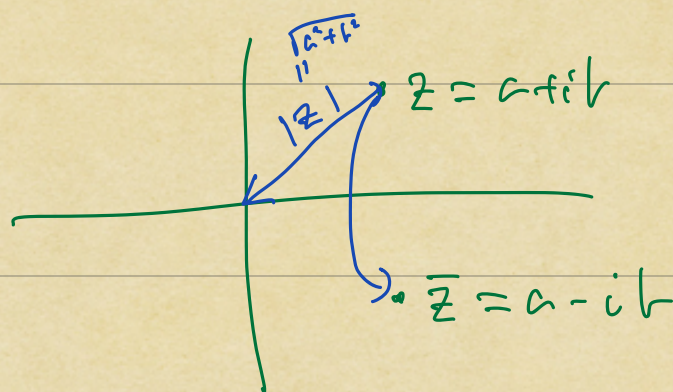
Defⁿ $z = a + ib \in \mathbb{C}$

The complex conjugate is

$$\overline{z} = a - ib \in \mathbb{C}$$

"zee-bar"

$a + i(-b)$



Fact: $z \cdot \overline{z} = (a + ib)(a - ib)$

$$= a^2 - (ib)^2$$

$$= a^2 + b^2$$

Defⁿ $|z| = \sqrt{z \cdot \overline{z}}$

"modulus or" absolute value $= \sqrt{a^2 + b^2} \in \mathbb{R}$

Example:

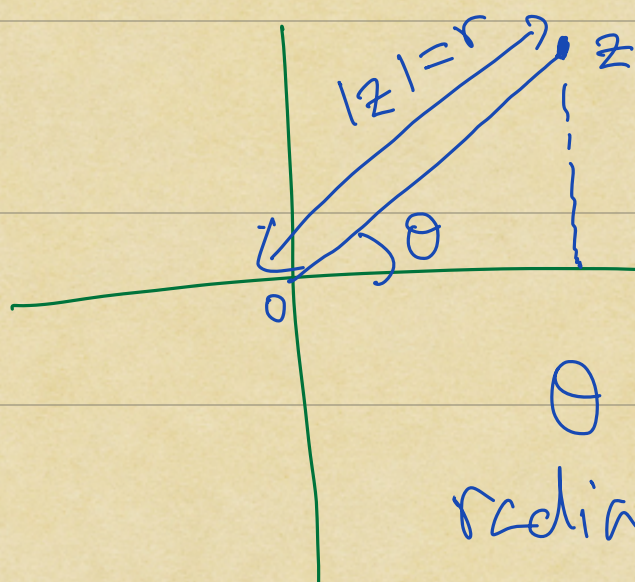
$$z = 3 + 5i$$

$$\bar{z} = 3 - 5i$$

$$|z|^2 = 3^2 + 5^2 = 34$$

$$\begin{aligned} \Rightarrow z^{-1} &= \frac{\bar{z}}{|z|^2} = \frac{3}{34} - \frac{5i}{34} \\ &= \frac{3}{34} - i \cdot \frac{5}{34} \end{aligned}$$

Euler's notation



$$\begin{aligned} z &= r e^{i\theta} \\ &= r e^{i\theta} \end{aligned}$$

θ measured in radians, $\theta \in [0, 2\pi)$

Examples:

$$(1) \quad 1 = 1 \cdot e^{i0}$$

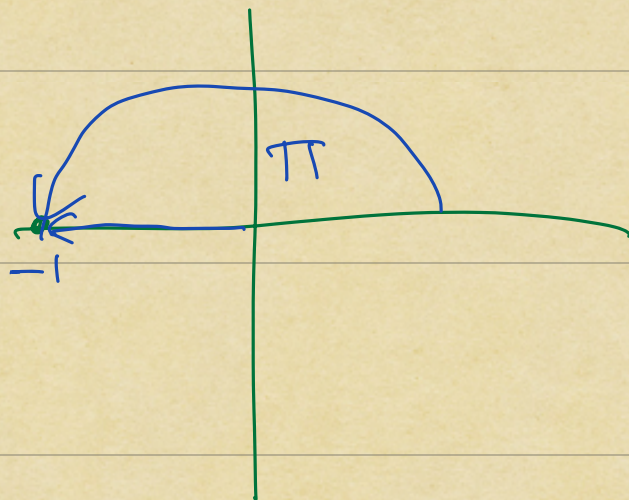
$$= 1 \cdot e^{i2\pi}$$

$$= 1 \cdot e^{2\pi i}$$

$$= 1 \cdot e^{2n\pi i} \quad n \in \mathbb{Z}.$$

$$(2) \quad -1 = 1 \cdot e^{i\pi}$$

$$= 1 \cdot e^{(2n+1)\pi i}, \quad n \in \mathbb{Z}$$



$$e^{i\pi} = -1$$